Optimal Spacecraft Attitude Control Using Linear Error Quaternion Closed-Loop Dynamics

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***Abstract*—In this report, the quaternion attitude parameter is used to turn the attitude control problem into an error-state regulation problem between current and desired orientation states. Quaternion kinematics and a multiplicative error quaternion are employed to formulate linear closed-loop dynamics. Next, a constant optimal feedback gain is selected by solving the continuous-time Algebraic Riccati Equation (ARE) in traditional Linear Quadratic Regulator (LQR) controller design. The proposed attitude controller was implemented into Basilisk, an open-source astrodynamics simulation framework, and successfully oriented a spacecraft along a desired attitude reference trajectory. Full-state feedback was provided by a stochastic estimator for Linear Quadratic Gaussian (LQG) implementation. The proposed optimal controller yielded lower cost when compared to a pole placement feedback gain and produced desirable performance when compared to a proportional derivative (PD) modified Rodrigues parameter (MRP) control law.**

I. Introduction

Precise attitude control is desirable across a wide variety of aerospace systems such as missiles, spacecraft, aircraft, and drones. Many methods have been proposed to track a desired orientation trajectory, each having their own advantages and disadvantages. Open-loop attitude control techniques require a predetermined trajectory, which is often undesirable for aerospace systems [1]. In all scenarios challenges arise due to highly non-linear governing dynamical equations, singularities in the selected attitude parameter, large slew maneuvers, incomplete state knowledge due to sensor availability, and the need for an optimal control law.

The goal of the attitude controller proposed in this work is to provide a robust, precise, and optimal attitude control law while approaching the problem in a simplified way without making assumptions or linearizations. To achieve this end first, the attitude of the body frame of the spacecraft is defined with respect to (w.r.t.) the Earth Centered Inertial (ECI) frame. This choice eliminates any need to calculate relative rotation rates between non-inertial frames. Secondly, the quaternion attitude parameter is chosen because of the linear governing kinematic equations. Next, the tracking problem is turned into a regulation problem by defining a multiplicative error quaternion that the designer aims to drive to zero. Following from the error quaternion definition linear closed-loop dynamics are assumed, which facilitates an optimal LQR gain calculation. Finally, the linear quaternion kinematics allow for the derivation of an analytical solution for applied torque into the system with arbitrary gains.

II. Methodology

It is important to define the notation used for the vector and scalar portion of the quaternion attitude parameter (1). The control law derived in this work is capable of controlling the orientation between any two reference frames. The quaternion that describes the orientation of frame B w.r.t. frame A can be expressed as, $\mathbf{q_{B/A}}$, along with all of it's time derivatives. The associated angular rate in the following derivation is the angular rate of frame B w.r.t. frame A resolved in the frame B, $\omega^B_{B/A}$, along with all of it's time derivatives.

$$\mathbf{q} = \begin{bmatrix} q_0 \\ Q \end{bmatrix} = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \tag{1}$$

For the remainder of the report the attitude, $\mathbf{q}$, and it's time derivatives will be referred to as the spacecraft body frame w.r.t. the ECI frame. The angular rate, ω, and it's time derivatives will be referred to as the angular rate of the spacecraft body frame w.r.t. the ECI frame resolved in the spacecraft body frame.

A. Error Quaternion Closed-Loop Dynamics

After the selection of the quaternion attitude parameter and corresponding reference frames has been made, it is important to analyze the governing linear kinematic equations, depicted in 2, 3, and 4 with the matrices $\mathbf{\Xi}(\mathbf{q})$ and $\mathbf{\Omega}(\boldsymbol{\omega})$ defined in (5) and (6). Note that $[\cdot]_\times$ represents the skew symmetric matrix of a vector.

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{\Omega}(\boldsymbol{\omega})\,\mathbf{q} = \frac{1}{2}\,\mathbf{\Xi}(\mathbf{q})\,\boldsymbol{\omega} \tag{2}$$

$$\ddot{\mathbf{q}} = \frac{1}{2}\Xi(\mathbf{q})\dot{\boldsymbol{\omega}} + \frac{1}{2}\mathbf{\Omega}(\boldsymbol{\omega})\dot{\mathbf{q}} \tag{3}$$

$$\dot{\boldsymbol{\omega}} = -\mathbf{J}^{-1}[\boldsymbol{\omega}]_\times\mathbf{J}\boldsymbol{\omega} + \mathbf{J}^{-1}\mathbf{L} \tag{4}$$

$$\mathbf{\Xi}(\mathbf{q}) = \begin{bmatrix} -Q^\top \\ q_0\mathbf{I}_{3\times3} + [Q]_\times \end{bmatrix} \tag{5}$$

$$\mathbf{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \tag{6}$$

From (4) we see that the applied torque, $\mathbf{L}$, is the necessary input to the spacecraft actuator and therefore must be apart of our control law. The only unknown in (4) is indeed $\mathbf{L}$ as the inertia matrix, $\mathbf{J}$, is known from mass analyses and the angular rate, ω, can be provided from onboard gyroscopes.

Let us simultaneously take into consideration that we desire to form a linear time-invariant (LTI) closed-loop state space dynamical model. Known optimal feedback gain calculation methods can then be employed. Our LTI closed-loop dynamics input, $\mathbf{u}$, must contain $\dot{\boldsymbol{\omega}}$ and consequentially $\mathbf{L}$.

At this point our goal is to form closed-loop quaternion LTI dynamics in which the control, $\mathbf{u}$, contains $\dot{\boldsymbol{\omega}}$. It is now advantageous to consider regulating error dynamics rather than solving the tracking problem. The multiplicative error quaternion between the current and desired attitude states is defined in (7). As the error is defined, it is important to note that $(\cdot)$ represents current state and $(\cdot)_d$ represents the desired state.

$$\delta\mathbf{q} = \mathbf{q} \otimes \mathbf{q}_d^{-1} \tag{7}$$

Note that $\otimes$ represents the multiplication of quaternions and is non-linear in nature. The vector portion of the error quaternion is linear and can be described in (8). From (8) and (1) if δQ approaches zero then the current quaternion attitude state approaches the desired quaternion state.

$$\delta Q = \mathbf{\Xi}^T(\mathbf{q}_d)\,\mathbf{q} \tag{8}$$

At this point it is convenient to assume the error quaternion vector to follow the prescribed linear closed-loop dynamics, described equivalently by (9) and (10). The associated LTI state-space formulation is given by (11).

$$\delta\ddot{Q} + K_2\delta\dot{Q} + K_1\delta Q = 0 \tag{9}$$

$$\delta\ddot{Q} = \mathbf{u} = -K\begin{bmatrix}\delta Q \\ \delta\dot{Q}\end{bmatrix} \tag{10}$$

$$\dot{\mathbf{x}} = \begin{bmatrix}\mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3}\end{bmatrix}\mathbf{x} + \begin{bmatrix}\mathbf{0}_{3\times3} \\ \mathbf{I}_{3\times3}\end{bmatrix}\mathbf{u} \tag{11}$$

Upon inspection of (10) we have closed-loop error quaternion dynamics with the input control being the second time derivative of the error quaternion. Notice that taking two time derivatives of (8) will yield an $\dot{\boldsymbol{\omega}}$ term, which consequentially contains $\mathbf{L}$, the applied torque into the system. Our goal of formulating closed-loop LTI dynamics in which the control, $\mathbf{u}$, contains $\dot{\boldsymbol{\omega}}$ is achieved. At this point the feedback gain may be selected with any method of choice.

B. LQR Feedback Gain

It is desirable for the feedback gain in the closed-loop control law to minimize the quadratic cost function (12) on an infinite time horizon. Note that Q and R are positive-definite symmetric matrices that penalize the state and control input respectively. Given the state is governed by LTI dynamics of the form (13), the optimal solution to quadratic cost minimization is found by solving the continuous-time ARE (14) and then solving for the constant feedback gain (15).

$$J = \int_0^\infty \left(\mathbf{x}^\top Q\mathbf{x} + \mathbf{u}^\top R\mathbf{u}\right)dt \tag{12}$$

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \tag{13}$$

$$A^\top P_\infty + P_\infty A - P_\infty BR^{-1}B^\top P_\infty + Q = 0 \tag{14}$$

$$K_\infty = R^{-1}B^\top P_\infty \tag{15}$$

Solving the Riccati equation on the infinite time horizon yields a constant optimal closed-loop feedback gain, which significantly reduces the complexity of real-time controller implementation. It is also important to note that if (A, C), where $Q = C^TC$, is observable and there exists a positive definite limiting solution to the continuous-time ARE, P_∞. The closed-loop plant, $A_{cl} = (A - BK_\infty)$ with constant gain K_∞ found using P_∞, is guaranteed to be stable [3]. A pole analysis of the proposed control law can be found in Section III-A.

C. Analytical Expression for Applied Torque

Our attention must be directed towards solving for the applied torque, $\mathbf{L}$, that achieves the closed-loop dynamics in (9). To achieve this end two time derivatives of δQ must be taken.

$$\delta Q = \mathbf{\Xi}^T(\mathbf{q}_d)\,\mathbf{q} \tag{16}$$

$$\delta\dot{Q} = \mathbf{\Xi}^\top(\dot{\mathbf{q}}_d)\,\mathbf{q} + \mathbf{\Xi}^\top(\mathbf{q}_d)\,\dot{\mathbf{q}} \tag{17}$$

$$\delta\ddot{Q} = \mathbf{\Xi}^\top(\ddot{\mathbf{q}}_d)\,\mathbf{q} + 2\mathbf{\Xi}^\top(\dot{\mathbf{q}}_d)\,\dot{\mathbf{q}} + \mathbf{\Xi}^\top(\mathbf{q}_d)\,\ddot{\mathbf{q}} \tag{18}$$

At this point (16), (17), and (18) must be substituted into (9) to yield (19).

$$\begin{aligned} &\mathbf{\Xi}^\top(\mathbf{q}_d)\,\ddot{\mathbf{q}} \\ &+ \left(2\mathbf{\Xi}^\top(\dot{\mathbf{q}}_d) + K_2\mathbf{\Xi}^\top(\mathbf{q}_d)\right)\dot{\mathbf{q}} \\ &+ \left(\mathbf{\Xi}^\top(\ddot{\mathbf{q}}_d) + K_2\mathbf{\Xi}^\top(\dot{\mathbf{q}}_d) + K_1\mathbf{\Xi}^\top(\mathbf{q}_d)\right)\mathbf{q} = 0 \end{aligned} \tag{19}$$

Let us take note that our desired applied torque, $\mathbf{L}$, arises in the $\ddot{\mathbf{q}}$ term and is displayed in (20). At this point (2) and (20) can be substituted into (19) and then solved for $\mathbf{L}$. This is an exercise left to the reader. The analytical expression for $\mathbf{L}$ is given by (21).

$$\ddot{\mathbf{q}} = -\frac{1}{2}\boldsymbol{\Xi}(\mathbf{q})\mathbf{J}^{-1}[\boldsymbol{\omega}]_{\times}\mathbf{J}\boldsymbol{\omega} - \frac{1}{4}(\boldsymbol{\omega}^T\boldsymbol{\omega})\mathbf{q} + \frac{1}{2}\mathbf{J}^{-1}\mathbf{L} \quad (20)$$

$$\boxed{\begin{aligned}\mathbf{L} = [\boldsymbol{\omega}]_{\times}\mathbf{J}\boldsymbol{\omega} + 2\mathbf{J}\left[\boldsymbol{\Xi}^{\top}(\mathbf{q}_d)\boldsymbol{\Xi}^{\top}(\mathbf{q})\right]^{-1} \\ + \left\{\frac{1}{4}\left(\boldsymbol{\omega}^{\top}\boldsymbol{\omega}\right)\mathbf{q}_d - \boldsymbol{\Xi}^{\top}(\dot{\mathbf{q}}_d)\boldsymbol{\Omega}(\boldsymbol{\omega}) - \boldsymbol{\Xi}^{\top}(\ddot{\mathbf{q}}_d)\right. \\ \left. -K_1\boldsymbol{\Xi}^{\top}(\mathbf{q}_d) - K_2\left[\frac{1}{2}\boldsymbol{\Xi}^{\top}(\mathbf{q}_d)\boldsymbol{\Omega}(\boldsymbol{\omega}) + \boldsymbol{\Xi}^{\top}(\dot{\mathbf{q}}_d)\right]\right\}\mathbf{q}\end{aligned}} \quad (21)$$

In the derivation of the error quaternion control law no lineariztion, small angle approximations, or any other precision degrading assumptions were made. It is important to note the attitude control law does have a singularity limitation. The inverse of $[\boldsymbol{\Xi}^{\top}(\mathbf{q}_d)\boldsymbol{\Xi}^{\top}(\mathbf{q})]$ will always exist for nonzero scalar portion of the error quaternion, $\delta q = \mathbf{q}_d^T\mathbf{q}$. δq is zero for rotations of $\pm 180°$ in the tracking error. Therefore, care must be taken when tracking errors reach $\pm 180°$. This is analogous to the non-uniqueness of the quaternion attitude representation, where $\mathbf{q}$ and $-\mathbf{q}$ describe the same physical rotation.

To implement the attitude control law (21) one must select $\mathbf{q}_d$, ω_d, and $\dot{\omega}_d$. The desired quaternion rate, $\dot{\mathbf{q}}_d$, and acceleration, $\ddot{\mathbf{q}}_d$, can then be found using (2) and (3). The current attitude, $\mathbf{q}$, can be obtained from an onboard stochastic estimator or observer, the angular rate, ω, can be provided by onboard gyroscopes, and the body's inertia, $\mathbf{J}$, is either estimated or known prior to controller implementation.

III. Simulation

A. Controller Implementation

In the simulation results to follow the ARE was solved using the SciPy Python package with the state and control penalty matrices given by $Q = 1 \times 10^{-4} I_{6\times 6}$ and $R = I_{3\times 3}$. This Q and R selection results in:

$$P_\infty = \begin{bmatrix}\sigma I_{3\times 3} & \rho I_{3\times 3} \\ \rho I_{3\times 3} & \tau I_{3\times 3}\end{bmatrix}$$

where $\sigma = 0.00141774, \quad \rho = 0.01, \quad \tau = 0.14177447$

which yields the gain $K_\infty = [K_1 K_2]$, $K_1 = 0.01 I_{3\times 3}$ and $K_2 = 0.14177 I_{3\times 3}$.

A pole analysis of the original system depicted in (11) yields six eigenvalues of 0 + 0j, which lie at the origin of the real and imaginary plane, leaving the stability of the system undetermined.

The poles of the closed-loop system given by $A_{cl} = (A - BK_\infty)$ are three sets of $s_{1,2} = -0.07088723 \pm 0.07053368j$. The resulting natural frequency and damping ratio of the closed-loop system are $\omega_n = 0.1$ and $\zeta = 0.709$. All its poles contain negative real portions, guaranteeing stability. This result does not come by surprise. P_∞ is positive definite and the observability matrix of the system (A, C) is full rank, n, as $x \in \mathbb{R}^n$. Therefore, as discussed in Section II-B, the closed-loop system is guaranteed to be stable. This theorem is a major benefit of the infinite-time horizon LQR gain implementation.

B. Simulation Environment

In all of the subsequent simulation scenarios the time derivative of desired angular rate, $\dot{\omega}_d$, was set to zero. The initial desired attitude, $\mathbf{q}_d$, and angular rate, ω_d, were set and $\mathbf{q}_d$ was propagated using (2).

The error quaternion closed-loop optimal attitude controller was verified using Basilisk [2], an astrodynamics simulation framework, where it successfully drove a low Earth orbit (LEO) spacecraft's orientation along various desired reference trajectories.

The simulated spacecraft mass properties are:

$$\mathbf{J} = \begin{bmatrix}30.0 & 10.0 & 5.0 \\ 10.0 & 20.0 & 3.0 \\ 5.0 & 3.0 & 15.0\end{bmatrix} \text{ kg}\cdot\text{m}^2, \quad m = 10.0 \text{ kg}$$

The spacecraft's orbit is given by:

$$\begin{aligned} \text{Semi-major axis:} \quad & a = r_{\text{LEO}} = 7,000km \\ \text{Eccentricity:} \quad & e = 0.0001 \\ \text{Inclination:} \quad & i = 0° \\ \text{RAAN:} \quad & \Omega = -90° \\ \text{Argument of Perigee:} \quad & \omega = 0° \\ \text{True Anomaly:} \quad & \nu = 0° \end{aligned}$$

The spacecraft attitude, $\mathbf{q}$, and angular rate , ω, were fed into the controller module by Basilisk's stochastic SimpleNav module for LQG controller implementation. The control input fed back to the spacecraft was a desired applied torque with no modeled actuator(s). At this point the reader is encouraged to remind themselves of the defined attitude and rotation rate described in Section II.

IV. Results and Discussion

A. Comparison with Pole Placement Gain

The first simulation scenario compares the proposed controllers LQR gain selection with a pole placement gain selection method to deomonstrate both the proposed attitude controllers functionality and optimality. The initial spacecraft orientation, $\mathbf{q}$, was aligned with the ECI frame with an initial tip-off rate of, $\omega = [0, 5, 0]^T deg/s$.

For the pole placement gain calculation, the natural frequency and damping ratio were selected as $\omega_n = 0.1$ and $\zeta = 0.9$. The SciPy Python package's pole placement function was then used to calculate the closed-loop feedback gain for the system.

The goal of this scenario is to point the LEO spacecraft's body z-axis toward Earth. To achieve this end the initial desired attitude is given by, $\mathbf{q}_d = [\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0, 0]^T$, and desired angular rate, $\omega_d = [0, 0.0068, 0]^T deg/s$. The results are depicted in Fig. 1 through 7.

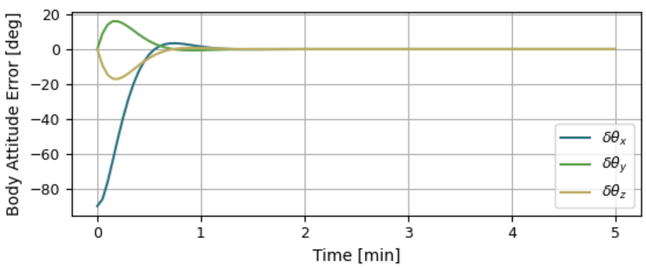

Fig. 1. Body Frame Attitude Error LQR Gain

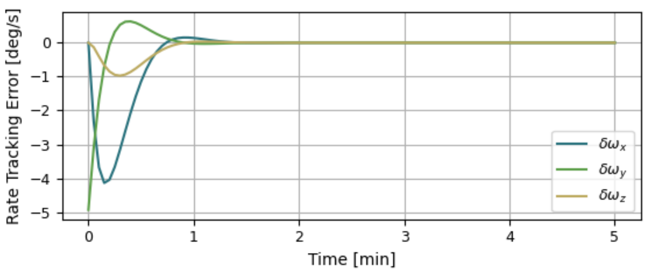

Fig. 2. Angular Rate Tracking Error LQR Gain

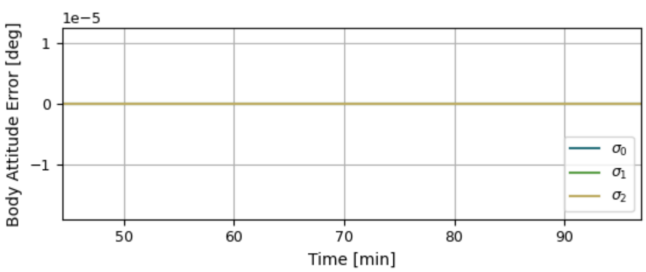

Fig. 3. One Orbit Fine Pointing LQR Gain

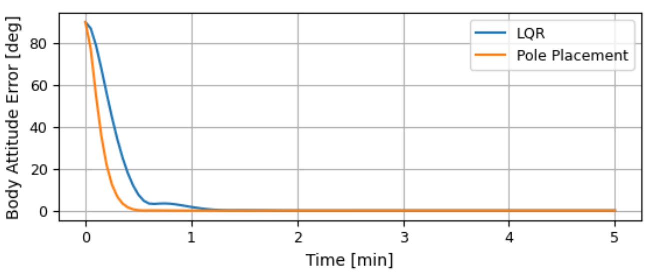

Fig. 4. Magnitude Body Frame Attitude Error Comparison

Fig. 1 and 2 are provided to show that all three axes individually converge to the desired orientation state. Fig. 3 is provided to show that the proposed attitude controller is able to achieve fine pointing for one orbital period. [1] makes the distinction that the time required for the damped oscillations in the spacecraft body y-axis attitude error should decay to within 2% of zero by $4/(\zeta\omega_n) = 56.4175$ seconds. Upon inspection

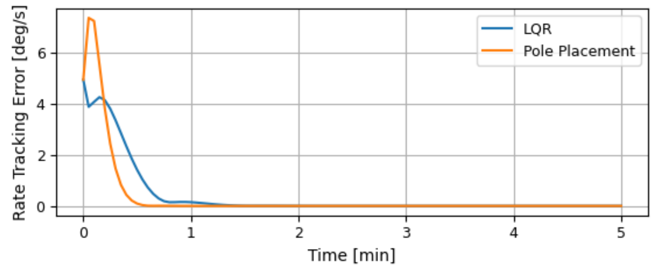

Fig. 5. Magnitude Angular Rate Tracking Error Comparison

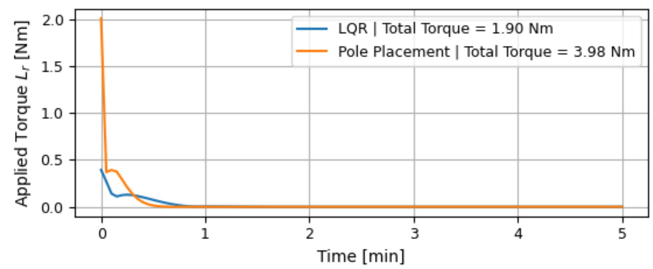

Fig. 6. Magnitude Applied Torque Comparison

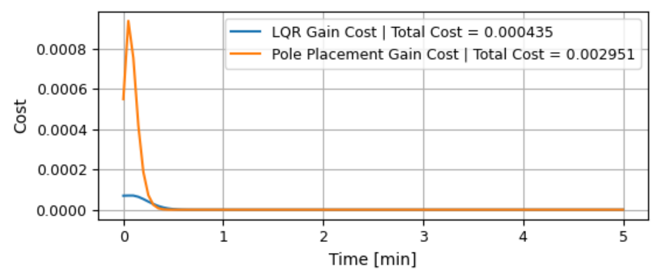

Fig. 7. Cost Comparison

of Fig. 1, the controller meets this settling time requirement, which alludes to a successful implementation.

Fig. 4 through 7 compare the performance of the optimal LQR gain to the pole placement gain. Fig. 4, 5, and 6 show that the pole placement feedback gain drives the system state error to zero more quickly. This is analogous to say that the pole placement gain implementation reduces state error more aggressively. This is consistent with our spring-mass-damper analysis where ω_n was the same for each system, but the ζ for the pole placement system was $\zeta = 0.9$ compared to $\zeta = 0.709$ for the optimal LQR gain system.

Even though the pole placement gain drives the state error to zero more quickly, is cannot guarantee optimality when held accountable by the LQR cost function. In Fig. 7 the discrete cost is plotted over time. The LQR gain implementation's total cost is $J^*_{total} = 0.000435$ compared to the pole placement's total cost, $J_{total} = 0.002951$.

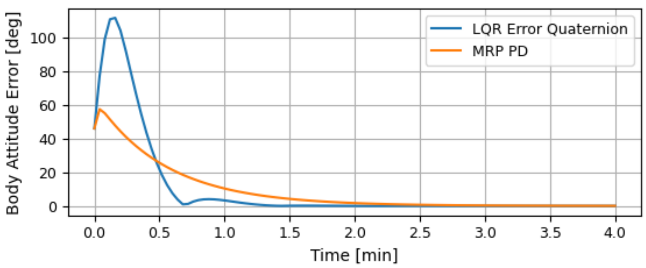

Fig. 8. Magnitude Body Frame Attitude Error Comparison to MRP PD

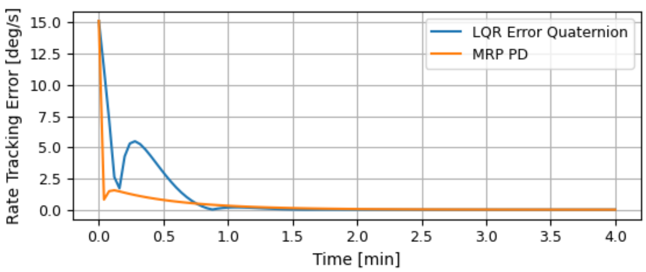

Fig. 9. Magnitude Angular Rate Tracking Error Comparison to MRP PD

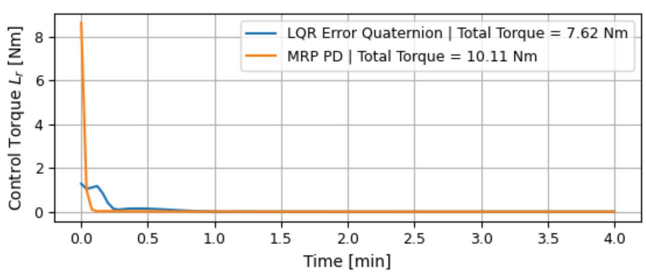

Fig. 10. Magnitude Applied Torque Comparison to MRP PD

B. Comparison with MRP PD Attitude Controller

The second scenario is provided to compare the proposed optimal closed-loop attitude controller with an entirely different control law and to show the proposed controller's ability to detumble amongst high initial attitude and rate error. The selected attitude control law of comparison is an MRP PD control law, which is a pre-existing Basilisk attitude control module.

The initial spacecraft orientation, $\mathbf{q}$, was computed with the principle Euler vector of $\theta = [0, 45, 10]^T deg$ and an initial tip-off rate of, $\omega = [15, 2, 0]^T deg/s$. The desired attitude state was a set to, $\mathbf{q}_d = [1, 0, 0, 0]^T$ and $\omega_\mathbf{d} = [0, 0, 0]^T deg/s$ for both controllers. The MRP controller's proportional gain was set to $K = 3.5$ and derivative gain to $P = 30$.

Fig. 8 and 9 show the proposed optimal controller's ability to drive high attitude and rate tracking error to zero. In comparision with the MRP PD controller, from Fig. 8 and 9 it is clear to see that the proposed attitude controller reduces state error more quickly. Upon inspection of Fig. 10, the proposed optimal attitude controller simultaneously requests less total applied torque, $L_r = 7.62$ Nm, compared to the MRP PD controller's, $L_r = 10.11$ Nm. Overall, the proposed attitude controller is able to reduce high state error more quickly while requesting less applied torque. Making it a more desirable option when compared to the MRP PD controller.

V. Conclusion

In conclusion, the proposed attitude control law simplifies the reference orientation tracking problem by selecting the quaternion attitude parameter, defining a multiplicative error-state to create a regulation problem, and prescribing closed-loop state feedback dynamics. Choosing to select the feedback gain with the LQR method guarantees optimality and closed-loop stability.

The proposed optimal attitude controller was implemented into the Basilisk astrodynamics simulation framework to control a LEO spacecraft's orientation. In an Earth pointing scenario, the proposed optimal controller successfully achieved fine pointing, while producing less cost when compared to a pole placement gain calculation method. When compared to an MRP PD control law, the proposed attitude controller was able to successfully detumble from high state error faster, while requesting less applied torque.

References

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